第一部分 自由泳的比赛规定

Part I: Freestyle competition rules

自由泳比赛中，可采用任何泳式。但在个人混合泳及混合泳接力比赛中，自由泳是指除蝶泳、仰泳、蛙泳以外的泳式。

Freestyle means that in an event so designated the swimmer may swim any style, except that in individual medley or medley relay events, freestyle means any style other than backstroke, breaststroke or butterfly.

每次转身和到达终点时，身体的某一部分必须触及池壁。

Some part of the swimmer must touch the wall upon completion of each length and at the finish.

在整个游程中，运动员身体的某一部分必须露出水面。在出发和转身时，允许运动员身体完全没入水中。出发和每次转身后，在 15 米前（含 15 米）运动员头的一部分必须露出水面。

Some part of the swimmer must break the surface of the water throughout the race, except it shall be permissible for the swimmer to be completely submerged during the turn and for a distance of not more than 15 metres after the start and each turn. By that point, the head must have broken the surface.

第二部分自由泳技术动作

Part II: Freestyle Techniques

爬泳？ Front Crawl？

爬泳的名称来自于它的外观特征。在游爬泳时，身体俯卧在水面，腿上下交替打水,两臂轮流划水，动作很像爬行，故被称为“爬泳”

The name of the Front Crawl comes from its appearance. During the Front Crawl, the body is lying on the water, the legs alternately hit the water, and the two arms alternately stroke the water. The movement is like crawling, so it is called "Front Crawl“

在现代竞技游泳比赛中，并没有“爬泳”这个项目，而设有“自由泳”项目。在自由泳比赛中运动员可以用任何泳姿游进。由于爬泳速度最快运动员几乎都用爬泳游进，故而爬泳也被称为“自由泳”

In the modern competitive swimming competition, there is no "Front Crawl" project, and there is a "freestyle" project. In the freestyle competition, athletes can swim in any stroke. Because the fastest crawling athletes almost all use the Front Crawl to swim in, the crawling is also known as "freestyle."

自由泳技术动作由身体姿势、腿部动作、臂部动作、呼吸及完整配合几部分动作通过协调的配合构成。

The freestyle technical action consists of a coordinated fit of body posture, leg movements, arm movements, breathing, and a complete set of actions.

一、身体姿势 Body posture

在游任何一种泳式时，理想的身体姿势是最大限度地减小阻力。为了达到这个目的,自由泳的身体姿势有以下特点:

The ideal body posture is to minimize drag when swimming in any of the swimming styles. In order to achieve this goal, the freestyle body posture has the following characteristics:

1.身体呈水平姿势 The body is in a horizontal position

基本姿势是身体尽量保持水平，身体位于水面上较高的位置以减小形状阻力。

The basic posture is that the body is kept as level as possible, and the body is located at a higher position on the water surface to reduce the shape resistance.

2.头部位置 Head position

正确的头部位置是与水面平行，眼望池底。这样颈部肌肉放松，手臂和身体都能够充分地伸展。

The correct head position is parallel to the water surface and the bottom of the pool. This way the neck muscles relax and the arms and body are fully extended.

较高的头部位置容易使躯干和腿下沉，从而使形状阻力增大。

A higher head position tends to sink the torso and legs, thereby increasing the shape resistance.

根据流体力学的实验研究，当身体俯卧于水面以一定的速度前进时，头位于水面上时身体所遇到的阻力要明显大于头浸入水中时。如果试图通过抬高头部来提高身体位置，其结果必定是事与愿违,腿部就会更加下沉，且身体会上下起伏。

According to the experimental study of fluid mechanics, when the body is lying on the surface of the water at a certain speed, the resistance encountered by the body when the head is on the water surface is significantly greater than when the head is immersed in the water. If you try to raise your body position by raising your head, the result must be counterproductive, your legs will sink more, and your body will rise and fall.

二、自由泳腿部技术动作 The flutter kick

自由泳腿部动作虽有一定的推进力，但主要起平衡作用。

Although the freestyle leg movement has a certain propulsive force, it mainly plays a balancing role.

自由泳腿部打水由向下打水和向上打水两部分交替构成，其中下打动作较为有力，上打相对放松一些。

The freestyle kick is composed of two parts: the downward and the upward. The downward is more powerful and the upward is relatively relaxed.

腿打水过程中两脚应稍向内扣，踝关节放松，以髋关节为轴，由大腿带动小腿和脚掌，两腿交替做鞭打动作，两脚尖上下最大幅度 30~40 厘米，膝关节最大屈度约 130%。

During the kick process, the two legs should be slightly buckled inward, the ankle joint should be relaxed, the hip joint should be used as the axis, the calf and the sole of the foot should be driven by the thigh, and the legs should be alternately whipped. The maximum amplitude of the two toes is 30~40 cm, and the knee joint is the largest. The degree of flexion is about 130%.

三、自由泳臂部动作 The armstroke

自由泳臂部动作是推动身体前进的主要动力，划水周期可分为：入水、划水（抓水、拉水、推水）、出水、空中移臂。

Freestyle arm movement is the main driving force for the body to advance. The stroke cycle can be divided into: enter、downsweep(catch、pulling water、upsweep) release、recovery.

入水：入水点在身体纵轴和肩关节的延长线之间。入水时手指自然伸直并拢，臂内旋使肘关节抬高处于最高点，掌心斜向外下方，使手首先触水，然后是前臂，最后是上臂自然插入水中。

The entry：entry point is between the longitudinal axis of the body and the extension of the shoulder joint. When entering the water, the fingers naturally straighten and close together. The inner rotation of the arm causes the elbow joint to rise at the highest point, and the palm is inclined outward and downward, so that the hand first touches the water, then the forearm, and finally the upper arm is naturally inserted into the water.

抓水：手入水和伸臂后，屈腕，使手掌朝向外下方,沿螺旋曲线抓水。从抓水开始,手臂开始有了向后的运动分量，即手臂的动作开始产生阻力推进力了。抓水的动作方向是向下、向外和向后。

catch water: After entering the water and stretching the arm, bend the wrist so that the palm is facing the outside and grab the water along the spiral curve. From the beginning of the catching of water, the arm begins to have a backward component of motion, that is, the action of the arm begins to generate resistance propulsion. The direction of action of catching water is downward, outward and backward.

拉水：根据拉水时手臂的主要运动方向，把拉水分为“下划”和“内划"两个紧紧相连的环节。
pulling water：According to the main direction of movement of the arm when pulling water, the pulling water is the two closely connected links of “downsweep " and “insweep".

“下划”开始时，手臂继续向下并稍向外划动，手掌稍内旋对着后下方，肘关节继续弯曲，使手和前臂逐渐加速向下划动。

At the beginning of the " downsweep ", the arm continues to slide downward and slightly outward, the palm is slightly rotated inward and downward, and the elbow joint continues to bend, causing the hand and forearm to gradually accelerate downward.

当手向下划至最低点时即转为“内划”，掌心转为朝着内后方，手从肩的垂直面外侧向内、向上、向后加速划至胸部下方接近或略超过身体中线处，肘关节弯曲至 90°~ I20°角。

When the hand is drawn down to the lowest point, it will change to " insweep ", the palm will turn toward the inner and rear, and the hand will accelerate from the outer side of the shoulder to the inner side, upward and backward to the lower part of the chest to approach or slightly exceed the midline of the body. At the point, the elbow joint is bent to an angle of 90° to I20°.

推水的前半部分，掌心转为朝外后方，手掌几乎直接由胸下向腰下划动，直至划近大腿侧下方。

upsweep：Pushing the first half of the water, the palm is turned to the outer rear, and the palm is swept almost directly from the chest to the lower part of the waist until it is drawn to the lower side of the thigh.

出水：手臂划水结束后应立即改变手的迎角,肘外旋，使小指朝上,掌心朝向大腿。这样可以使手轻松地离开水面，避免阻力；出水的顺序是肩、上臂、前臂和手。

Release：immediately after the end of the stroke, the angle of attack of the hand should be changed, and the elbow should be turned outwards so that the little finger is facing upwards and the palm is facing the thigh. This allows the hand to easily get out of the water and avoid drag.；The order of the water is the shoulder, upper arm, forearm and hand.

移臂：高肘移臂-从出水开始肘关节就已经屈曲,随肩和上臂向前上方移动。当手越过肩关节时开始前伸，手臂的动作转为向前、向内和向下。

The recovery：High elbow arm -The elbow joint has flexed from the beginning of the water, moving forward and upward with the shoulder and upper arm. When the hand crosses the shoulder joint, it begins to extend forward, and the movement of the arm turns forward, inward, and downward.

直臂移臂--当手出水时，手臂几乎完全伸直，手在肘上直臂向前、向上、向外移动,肩关节外旋。

Straight arm - When the hand is out of the water, the arm is almost completely straightened, and the hand moves forward, upwards, and outwards on the elbow, and the shoulder joint is externally rotated.

四、两臂配合 Timing of the arms

两臂的配合有三种基本形式即前交叉配合、中交叉配合和后交叉配合。

There are three basic forms of the two arms, namely the front cross fit, the middle cross fit and the rear cross fit.

前交叉配合指一臂入水时另一臂在肩前方，与水平面成锐角也被称为迫逐式配合。（适合初学者）

Front cross fit：When one arm enters the water, the other arm is in front of the shoulder, and an acute angle with the horizontal plane is also called an forced cooperation. (suitable for beginners) 中交叉配合指一臂入水时另一臂划至肩下。

Middle cross fit：When one arm enters the water, the other arm is drawn to the shoulder.

后交叉配合指臂入水时另一臂划至腹下。

Rear cross fit：When the arm enters the water, the other arm is drawn to the abdomen.

五、呼吸模式 Breathing mode 每划水两次呼吸一

次 Breathe twice for each stroke

优点:增加每一游程呼吸次数，减少肌肉产生乳酸的概率，从而使你能够游更远的距离。

Advantages: Increase the number of breaths per run and reduce the probability of muscles producing lactic acid, allowing you to travel farther. 缺点:这种呼吸模式很容易导致身体向一侧歪斜而失去平衡。

Disadvantages: This breathing pattern can easily cause the body to skew to one side and lose balance.

每划水三次呼吸一次 Breathe once every three strokes 优点:划水动作可以更加平

衡 Advantages: the stroke can be more balanced

缺点:无法像每划水两次呼吸的模式那样吸入更多的氧气。

Disadvantages: It is not possible to inhale more oxygen as in the pattern of two strokes per stroke.

六、完整配合动作 Coordinating action

自由泳中臂、腿、呼吸协调一致的配合，是保持游进速度均匀性的基本条件。自由泳一般采用 6:2: 1 的配合技术，即在一个完整动作周期中，打腿 6 次(左、右腿各 3 次)，划水 2 次(左、右臂各 1 次)，呼吸 1 次。

The coordinated cooperation of arms, legs and breathing in freestyle is the basic condition for maintaining the uniformity of swimming speed. Freestyle generally uses 6:2: 1 matching technique, that is, 6 times in a complete action cycle (3 times in the left and right legs), 2 strokes (1 time each for the left and right arms), Breath 1 Times.